

Luar Shui Song

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EDUCATION

National University of Singapore <i>Master of Science in Computer Science</i>	Singapore Aug. 2020 – Oct. 2022
Nanyang Technological University <i>Bachelor of Engineering in Electrical & Electronic Engineering (cGPA: 4.78/5.00)</i>	Singapore Aug. 2016 – Aug. 2020

EXPERIENCE

Founding Engineer <i>Stealth Startup</i>	Jun. 2024 – Present Singapore
- Built production RAG system processing documents from internal and public sources, improving answer relevance by 20% through hybrid retrieval architecture on Postgres with pgvector - Designed and deployed ML evaluation framework enabling systematic A/B testing that informed multiple successful model improvements. - Built multi-agent research system orchestrating specialized LLM agents (search, analysis, synthesis) across web APIs, vector databases, and internal docs to generate comprehensive domain-specific research.	
AI Engineer <i>Espressif Systems</i>	Sep. 2022 – May. 2024 Singapore
- Implemented ML algorithms (GNN, RL, BO) for chip design, saving 20% time as compared to manual baseline. - Managed 8-GPU cluster to perform parallel training, model management and hyperparameter optimisation.	
Graduate Research Assistant <i>National University of Singapore</i>	Aug. 2020 – Aug. 2022 Singapore
- Applied deep reinforcement learning to real-world robotic applications outperforming classical control baselines. - Translated high-dimensional time-series tactile sensor data to useful insights via feature engineering. - Implemented high accuracy classification models using machine learning. - Published paper in IROS 2021 that achieved the Best Paper Award.	
Research, Design and Development Intern <i>Dyson</i>	May 2018 – Jul. 2018 Singapore
- Designed automation tests for position tracking, reduced 75% time in manual monitoring. - Utilised VICON with LabVIEW for 3D Pose, and the Python SDK for AWS IoT MQTT protocol.	

PROJECTS

Open Source Contributor at OpenROAD Project <i>Python, Tcl, LLMs</i>	May. 2023 – Present
- Conceptualised and open-sourced a proof-of-concept for a Large Language Model chatbot assistant. - Key maintainer for AutoTuner, an automatic hyperparameter optimizer framework for chip design. - Sole maintainer for OR-MCP, a MCP server for real-time AI-driven debugging for chip designers.	
Semantic Segmentation of Articulated Objects <i>Python, PyTorch</i>	Aug. 2019 – Dec. 2019
- Perform accurate joint (revolute, prismatic) classification in articulated objects with transfer learning. - Fine-tuned a pre-trained network for semantic segmentation with 0.931 <i>mIOU</i>.	
Emotion Analysis in Schizophrenia <i>MATLAB</i>	Jan. 2017 – Apr. 2017
Studied emotion classification (via Affectiva SDK) in schizophrenia using supervised classical machine learning techniques, such as SVM/SVR, kNN, CART, LR.	

SKILLS AND FRAMEWORKS

Languages: Python, JavaScript, SQL
ML & AI: PyTorch, TensorFlow, Scikit-Learn, HuggingFace, LangChain, RAG
Data & Infrastructure: Spark, Pandas, PostgreSQL, pgvector, MLFlow, Docker
Cloud & DevOps: AWS/GCP, Nomad, Terraform, Ansible, GitHub Actions
Spoken Languages: English, Chinese (Mandarin)